☁️ CLOUD INFRASTRUCTURE DELIVERY HEALTH DASHBOARD

Role: Technical Lead / Delivery Manager **Tool:** Copilot in Excel (VBA Macro Generation) **Duration:** 30 min · Level 3 Workshop Challenge

CHALLENGE OVERVIEW

You are a **Technical Lead or Delivery Manager** in a Cloud Business Group responsible for tracking delivery health across three active sprints. You have been given a three-sheet Excel workbook containing real operational sprint data. Your challenge is to **use Copilot in Excel to write VBA macros** that transform raw data into professional dashboard visualisations — without writing a single line of code manually.

Workbook: CLOUD_Sprint_Health_Dashboard_Data.xlsx

Sheet	Contents	Rows	Key Fields
Sprint Backlog	60 cloud infrastructure tickets across Sprints 12–14, 4 squads, 10 epics	60	Ticket ID (CLOUD-NNN), Status, Assignee, Squad, Original / Current Points, Blocker, Week Number
Squad Targets	Planned vs actual velocity and blocker budgets for each squad per sprint	12	Planned Velocity, Actual Velocity, On-Time Target %, Blocker Count, Health Status
Sprint Ceremonies	Sprint events from Planning through Retrospective for Sprints 12–14	14	Ceremony type, Facilitator, Key Decisions, Risks Raised, Status

Squads: Infrastructure · Platform Engineering · Security & Compliance · Cloud Migration

Epics: AWS Migration Runway · Kubernetes Platform Uplift · FinOps Cost Optimisation · Zero Trust Security Framework · CI/CD Pipeline Modernisation · Disaster Recovery Assurance · Observability & Monitoring Stack · Multi-Cloud Networking · IAM & Access Governance · Container Registry Consolidation

TEST DATA FILE

📁 **CLOUD_Sprint_Health_Dashboard_Data.xlsx** — Three-sheet workbook containing 60 CLOUD-prefixed sprint tickets, squad velocity targets, and sprint ceremony notes across Sprints 12–14, pre-formatted with an amber facilitator notes row on each sheet.

WHY THIS CHALLENGE?

- **Manual status aggregation wastes delivery manager time.** Cloud delivery teams produce 60+ tickets per programme increment; tracking completion rates, blocker counts, and velocity trends across four squads without automation leads to reporting delays and dashboard inaccuracies that obscure programme risk until it is too late to remediate.
- **Blocked tickets in cloud programmes have outsized impact.** AWS quota limits, CAB scheduling, vendor provisioning delays, and cross-team IAM dependencies cause cascading blockers. A Technical Lead without a real-time blocker register has no systemic way to escalate the right issues to programme leadership before sprint goals are missed.
- **Sprint health patterns must be visible in seconds, not minutes.** Sprint 14 is at 28% completion with 5 live blockers — but this is buried across three sheets of raw data. Copilot-generated dashboards enable the Delivery Manager to walk into a steering committee with a single-screen health view, RAG-status scorecards, and burndown evidence — without manually constructing a new deck each sprint.

SUCCESS CRITERIA

- **The macro runs end-to-end without errors** and creates or refreshes a 'Dashboard' sheet each time it is executed, even when run a second time on the same workbook (re-runnability).

- **All chart data is drawn directly from the workbook sheets** — no hardcoded values — so that updating a ticket status in Sprint Backlog is automatically reflected in the dashboard on the next macro run.
- **The visualisation answers a real programme management question** within ten seconds of opening the dashboard: the participant can immediately identify Sprint 14 as the critical risk sprint, name the most-blocked squad, and locate the tickets requiring Technical Lead escalation.

HOW TO TEST

Step 1 — Upload the workbook:
Open Excel on your desktop or via Excel Online. Open **CLOUD_Sprint_Health_Dashboard_Data.xlsx**. Open Copilot from the Home ribbon (**Home** → **Copilot**). Ensure you are signed into your Microsoft 365 account.

Step 2 — Paste the prompt:
Copy the prompt from your chosen difficulty level (below). Paste it exactly into the Copilot chat panel and press Enter.

Step 3 — Insert the VBA macro:
When Copilot returns VBA code, click **Insert Code** or copy it manually. Open the VBA Editor (**Alt + F11**), paste into a new module (**Insert** → **Module**), then press **F5** or click Run.

Step 4 — Validate the output:
Compare your Dashboard sheet against the Expected Response section for your level. Check chart type, data accuracy, colour coding, and layout. Run the macro a second time to confirm re-runnability (no duplicate sheets or errors).

LEVEL 1 · EASY · Sprint Status Overview Pie Chart

What makes it Easy: single sheet read, straightforward COUNT logic, one chart type, fixed colour mapping — ideal first macro

COPILOT PROMPT (paste exactly as shown)

```
Using the 'Sprint Backlog' sheet in this workbook, write a VBA macro that does the
following:

1. Reads the 'Status' column (column C) and counts tickets for each of these five
statuses:
   Done | In Progress | Blocked | Supervisor Review | To Do

2. Creates a new sheet called 'Dashboard' (delete and recreate if it already
exists).

3. Writes a summary table in the Dashboard sheet starting at cell A1 with headers:
   Status | Count | Percentage

4. Creates a Pie Chart from that summary table with:
   - Title: "Sprint Status Overview — Cloud Infrastructure Programme"
   - Slice colours: Done = Green (#00B050), In Progress = Blue (#4472C4),
     Blocked = Red (#FF0000), Supervisor Review = Orange (#FFC000), To Do = Grey
(#808080)
   - Data labels showing both category name and percentage
   - Chart positioned at cell D1, sized 400 × 300 pixels

5. The macro must be re-runnable: if 'Dashboard' already exists, delete it before
recreating.
```

Expected Response

Copilot should generate a macro of approximately 50–70 lines that loops through column C of the Sprint Backlog sheet, uses a Select Case or If/ElseIf block to categorise each ticket, writes the summary table, and creates a pie

chart via the **Charts.Add** method. The macro should explicitly set each slice colour using the **SeriesCollection(1).Points(i).Interior.Color** property with RGB values. The expected summary table output is:

Status	Count	Percentage	Slice Colour
Done	25	41.7%	Green (#00B050)
In Progress	15	25.0%	Blue (#4472C4)
Blocked	9	15.0%	Red (#FF0000)
Supervisor Review	7	11.7%	Orange (#FFC000)
To Do	4	6.7%	Grey (#808080)
[object Object]	[object Object]	[object Object]	—

⚠️ **Common trap:** Participants may use COUNTIF on column C without anchoring the header row, counting the word 'Status' itself. The correct range to count is C3:C62 (rows 2 is the header, row 1 is the facilitator note). Confirm the total equals exactly 60.

LEVEL 2 · INTERMEDIATE · Velocity Stacked Bar & Burndown Chart

What makes it Intermediate: cross-sheet data aggregation, two chart types on one Dashboard, conditional formatting logic, dual-series burndown

📋 COPILOT PROMPT (paste exactly as shown)

```
Using the 'Sprint Backlog' and 'Squad Targets' sheets, write a VBA macro that:

SECTION A — STACKED BAR CHART (Sprint Velocity)
1. From the 'Squad Targets' sheet, aggregate per sprint:
   - Planned Velocity (sum of all 4 squad values)
   - Actual Velocity (sum of all 4 squad values)
   Compute Completion % = Actual / Planned * 100 (round to 1 decimal).

2. Write a Sprint Summary table to Dashboard!A1:D4 with headers:
   Sprint | Planned Velocity (pts) | Actual Velocity (pts) | Completion %

3. Create a Stacked Bar Chart from this table showing Actual (blue) and
   Remaining = Planned - Actual (light grey) stacked to the Planned total.
   Title: "Sprint Velocity — Cloud Infrastructure Programme"
   Position: Dashboard!F1, sized 500 × 280 pixels.

4. In the Sprint Summary table, highlight the Completion % cell RED (#FFC7CE
   fill, #9C0006 font) for any sprint below 70%.

SECTION B — BURNDOWN CHART (Weeks 0-6)
5. From 'Sprint Backlog', calculate remaining points per week end:
   Week 0 = SUM of all Original Points
   Week N = Week (N-1) - SUM of Original Points for Done tickets with Week Number =
N

6. Compute the Ideal burndown: Week 0 value / 6, decreasing linearly to 0.

7. Write a Burndown table to Dashboard!A7:C14 with headers:
   Week | Actual Remaining | Ideal Remaining

8. Create a Dual-Line chart from this burndown table.
   Actual line: solid orange. Ideal line: dashed grey.
   Title: "Programme Burndown — Sprints 12-14"
   Position: Dashboard!F11, sized 500 × 280 pixels.

The macro must be re-runnable.
```

Expected Response

Copilot should produce a macro of approximately 120–160 lines split across two logical blocks. Section A reads the Squad Targets sheet and sums velocity figures per sprint using a loop or SUMIF-equivalent in VBA. Section B iterates over Week Numbers 1–6 in the Sprint Backlog, summing **Original Points** for Done-status tickets at each week boundary to compute the actual burn. The expected table outputs are:

Sprint Summary Table

Sprint	Planned Velocity (pts)	Actual Velocity (pts)	Completion %
Sprint 12	102	95	93.1%
Sprint 13	100	73	73.0%
Sprint 14	100	30	30.0%

Burndown Table

Week	Actual Remaining (pts)	Ideal Remaining (pts)
0 (Start)	338	338
1 (Wk 1)	306	282
2 (Wk 2)	273	225
3 (Wk 3)	250	169
4 (Wk 4)	221	113
5 (Wk 5)	215	56
6 (Wk 6)	200	0

⚠️ **Common traps:**

- **Current Points vs Original Points:** The burndown must use Original Points (column I) for all tickets, then subtract the Original Points of Done tickets week by week — not Current Points. Using Current Points understates the total baseline.
- **Sprint 14 completion = 30.0%** (100 planned, 30 actual) — this is the only sprint that triggers the red conditional highlight. Ensure the threshold is strict: < 70%, not ≤ 70%. Sprint 13 at 73.0% must remain unformatted.
- **Burndown Week 0 = 338:** Total original points across all 60 tickets: Sprint 12 = 102, Sprint 13 = 120, Sprint 14 = 116. Participants who only sum a single sprint will get a truncated baseline and a misleading chart.

LEVEL 3 · DIFFICULT · Full Multi-Section Programme Health Dashboard

What makes it Difficult: five discrete dashboard sections, radar chart with normalised metrics, traffic-light scoring, filtered risk log, ceremony parsing — all in one re-runnable macro

📋 COPILOT PROMPT (paste exactly as shown)

```
Using all three sheets, write a VBA macro that builds a full Programme Health
Dashboard
on a 'Dashboard' sheet (delete and recreate if it already exists). The dashboard
must
contain exactly five sections laid out top-to-bottom:

SECTION 1 — SUMMARY BANNER (rows 1-4)
Display 4 KPI cells side by side: Total Tickets (60), % Complete (41.7%),
Active Blockers (9), Sprints At Risk (1 — Sprint 14 only).
Apply a dark navy (#1F2D3D) background with white bold text to each KPI label cell.
Apply a light blue (#DAEEF3) fill to each KPI value cell.

SECTION 2 — SQUAD RADAR CHART (rows 6-26)
Using 'Squad Targets' for the MOST RECENT SPRINT (Sprint 14), calculate 5 metrics
for each of the 4 squads and normalise each metric to a 0-1 scale
(divide by the maximum value across all squads for that metric):
  a) Velocity Score   = Actual Velocity / Planned Velocity
  b) On-Time Score    = Actual On-Time % / On-Time Target %
  c) Blocker Resistance = 1 - (Blocker Count / Max Blockers Allowed)  ← INVERTED
  d) Completion Score = Actual Velocity / MAX(Planned Velocity across all squads)
  e) Health Index     = IF(Health Status = "Green",1, IF("Amber",0.5, 0))
Normalise each metric column so the highest squad = 1.
Create a Radar Chart titled "Squad Health Radar — Sprint 14".
Position at Dashboard!G6, size 420 × 300 pixels.

SECTION 3 — SPRINT SCORECARD (rows 28-34)
Build a scorecard table: Sprint | Overall Completion % | Blocker Count | Health RAG
Colour the Health RAG cell: Green (#C6EFCE) = ≥80%, Amber (#FFEB9C) = 70-79%,
Red (#FFC7CE) = <70%.

SECTION 4 — RISK LOG (rows 36-48)
From 'Sprint Backlog', extract ALL tickets with Status = "Blocked".
Display: Ticket ID | Summary (truncated to 60 chars) | Squad | Sprint | Priority |
Blocker Reason
Sort by Sprint descending (Sprint 14 first), then Priority (Critical before High).
Apply a red fill (#FFC7CE) to Critical rows, orange (#FFE0B2) to High rows.

SECTION 5 — CEREMONY INSIGHTS (rows 50-65)
From 'Sprint Ceremonies', filter to Sprint 14 rows ONLY.
Display: Ceremony | Date | Key Decisions / Actions | Risks Raised | Status
Title the section "Sprint 14 Ceremony Insights — Technical Lead Escalations".
Apply amber (#FFEB9C) fill to any row where the Risks Raised cell is not blank.

All section titles must use bold navy text. The macro must be fully re-runnable.
```

Expected Response

Copilot should produce a macro of approximately 250–350 lines organised into clearly labelled Sub sections or a single Main Sub with helper calls. The expected outputs for each section are detailed below:

Section 1 — Summary Banner

KPI Label	Expected	Label Fill	Value Fill

	Value		
Total Tickets	60	Navy (#1F2D3D) + white text	Light Blue (#DAEEF3)
% Complete	41.7%	Navy (#1F2D3D) + white text	Light Blue (#DAEEF3)
Active Blockers	9	Navy (#1F2D3D) + white text	Light Blue (#DAEEF3)
Sprints At Risk	1	Navy (#1F2D3D) + white text	Light Blue (#DAEEF3)

Section 2 — Squad Radar Chart (Sprint 14 normalised metrics)

Squad	Velocity Score	On-Time Score	Blocker Resistance	Completion Score	Health Index
Infrastructure	0.29	0.49	0.33	0.29	0.00
Platform Engineering	0.46	0.55	0.33	0.46	0.00
Security & Compliance	0.27	0.38	0.67	0.27	0.00
Cloud Migration	0.19	0.33	0.00	0.19	0.00

⚠️ **Critical trap:** Blocker Resistance must be INVERTED: 1 – (Blocker Count / Max Blockers Allowed). Cloud Migration Squad has 4 blockers against a max of 3, yielding a negative raw score — this should be clamped to 0 before normalisation. Participants who simply use Blocker Count directly will produce a radar chart where the worst squad appears strongest.

Section 3 — Sprint Scorecard

Sprint	Overall Completion %	Total Blocker Count	Health RAG
Sprint 12	93.1%	1	Green ≥ 80%
Sprint 13	73.0%	6	Amber 70–79%
Sprint 14	30.0%	14	Red < 70%

Note: Blocker Count is cumulative (Sprint 12 = 1, Sprint 13 = 3 new + 3 carried forward = 6, Sprint 14 = 5 new + 9 unresolved = 14). Participants using only current-sprint Blocked ticket count will undercount Sprint 13 and 14.

Section 4 — Risk Log (all 9 Blocked tickets, Sprint 14 first)

Ticket ID	Squad	Sprint	Priority	Blocker Reason (summary)
CLOUD-049	Infrastructure	Sprint 14	Critical	ECR cross-account IAM trust pending Legal sign-off
CLOUD-051	Security & Compliance	Sprint 14	Critical	AD sync incomplete; GDPR data residency mapping blocked
CLOUD-052	Cloud Migration	Sprint 14	Critical	Vendor not confirmed Amazon Linux 2023 compatibility
CLOUD-050	Platform Engineering	Sprint 14	High	Datadog API key provisioning delayed – vendor procurement
CLOUD-053	Infrastructure	Sprint 14	High	AWS quota increase for us-east-1 pending; P1 case raised
CLOUD-036	Cloud Migration	Sprint 13	Critical	Data Analytics schema sign-off outstanding
CLOUD-037	Infrastructure	Sprint 13	Critical	CAB approval window not scheduled for DR DB replication
CLOUD-038	Platform Engineering	Sprint 13	High	NAT Gateway change pending firewall approval
CLOUD-018	Platform Engineering	Sprint 12	Critical	AWS Service Limit increase pending – Enterprise Support

Section 5 — Sprint 14 Ceremony Insights (Technical Lead Escalations)

Ceremony	Date	Key Decisions / Actions (summary)	Risks Raised	Status
Sprint Planning	03-Feb-25	5 carry-over blockers escalated by Technical Lead to Programme Director; Wave 3 assessment added as must-deliver	Resource gap confirmed – request declined	In Progress
Daily Standup Wk5	07-Feb-25	Only 4 tickets Done; Technical Lead raised P1 incident for CLOUD-049 with AWS TAM	Datadog vendor delay unresolved	In Progress
Mid-Sprint Review	10-Feb-25	Technical Lead flagged programme-level risk of Wave 3 slip to Delivery Manager; scope reduction proposed	GDPR blocker may breach compliance deadline	In Progress
Sprint Review	14-Feb-25	Pending – Sprint 14 still in progress	TBC	Planned

⚠️ **Common traps:**

- **Ceremony filter:** Only Sprint 14 rows (rows 12–15 of Sprint Ceremonies) must appear in Section 5. Participants who render all 14 ceremony rows fail the layout criteria.
- **Amber risk highlight:** Only rows with a non-empty Risks Raised cell should receive amber fill — the Sprint Review row ('TBC') should be treated as having content, so all four Sprint 14 rows will be amber-highlighted.
- **Role naming:** The facilitator in Sprint 14 ceremonies is 'Technical Lead' throughout. Macro output that references generic 'PM' or 'Project Manager' in ceremony note text indicates the participant did not adapt the prompt to the CLOUD role context.

SCORING RUBRIC

Criteria	Weight	Easy (1×, 10 pts max)	Intermediate (2×, 20 pts max)	Difficult (3×, 30 pts max)
Code runs without errors	30%	3 pts	6 pts	9 pts
Correct data & calculations	25%	2.5 pts	5 pts	7.5 pts
Chart type & formatting	20%	2 pts	4 pts	6 pts
Layout & readability	15%	1.5 pts	3 pts	4.5 pts
Re-runnability (no errors on 2nd run)	10%	1 pt	2 pts	3 pts
[object Object]	[object Object]	[object Object]	[object Object]	[object Object]

Total maximum score across all three levels: 60 points. Participants are encouraged to attempt levels in order. Partial credit is awarded by criteria row. Facilitators should award full marks only when the Dashboard sheet is visually clean, all data values match the expected outputs in this card, and the macro runs twice without error.